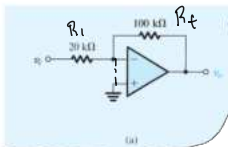
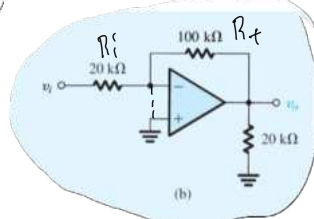


1. Assuming ideal op amps, find the voltage gain  $v_o/v_i$  and input resistance  $R_{in}$  of each of the circuits below:



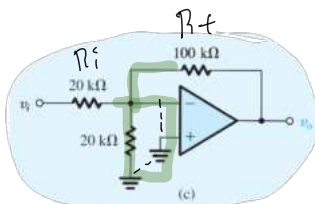
$$A = -\frac{R_f}{R_i} = -\frac{V_o}{V_i}$$

$$A = -\frac{100k}{20k} = -5$$

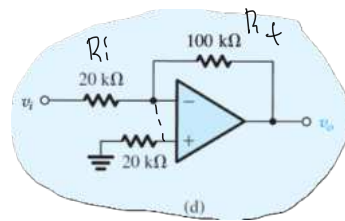


$$A = -\frac{R_f}{R_i}$$

$$A = -5$$

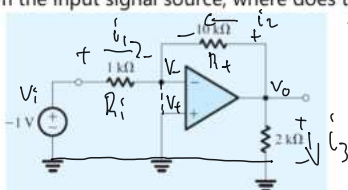


$$A = -\frac{R_f}{R_i} = -5$$



$$A = -\frac{R_f}{R_i} = -5$$

2. For the circuit below, assuming an ideal op amp, find the currents through all branches and the voltages at all nodes. Since the current supplied by the op amp is greater than the current drawn from the input signal source, where does the additional current come from?



$$V_- = V_+ = 0 \text{ [V]}$$

$$i_2 = \frac{V_o - V_-}{10k} = \frac{10}{10k} = 1 \text{ [mA]}$$

$$i_3 = \frac{V_o}{2k} = \frac{10}{2k} = 5 \text{ [mA]}$$

$$\frac{V_o}{V_i} = -\frac{R_f}{R_i}$$

$$i_1 = \frac{V_i - V_-}{R_i}$$

$$V_o = -\frac{10k}{1k} V_i$$

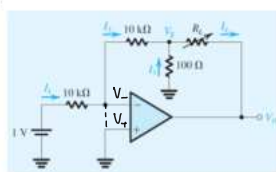
$$i_1 = \frac{-1}{1k} = -1 \text{ [mA]}$$

$$V_o = 10 \text{ [V]}$$

1 mA from source  
4 mA from op-amp output  
"extra" current must come from op-amp power source  
( $+V_{CC}$ ,  $-V_{EE}$ )

3. The circuit in below utilizes an ideal op amp.

- (a) Find  $I_1$ ,  $I_2$ ,  $I_3$ ,  $I_L$ , and  $V_x$ .  
(b) If  $V_o$  is not to be lower than  $-13 \text{ V}$ , find the maximum  
(c) allowed value for  $R_L$ .  
(d) If  $R_L$  is varied in the range  $100 \Omega$  to  $1 \text{ k}\Omega$ , what is the corresponding change in  $I_L$  and in  $V_o$ ?



$$V_- = V_+ = 0$$

$$\text{KCL at } V_-: -\frac{1}{10k} + \frac{-V_x}{10k} = 0$$

$$V_x = -1 \text{ [V]}$$

$$\text{at } V_x: I_2 + I_3 = I_L$$

$$0.1 \text{ m} + 10 \text{ m} = I_L$$

$$I_L = 10.1 \text{ [mA]}$$

$$a) I_1 = \frac{1 - V_-}{10k} = \frac{1}{10k} = 0.1 \text{ [mA]}$$

$$I_2 = \frac{V_- - V_o}{10k} = -\frac{V_x}{10k} = -\frac{-1}{10k} = 0.1 \text{ [mA]}$$

$$I_3 = \frac{0 - V_x}{100} = \frac{-(-1)}{100} = 10 \text{ [mA]}$$

$$d) I_L = I_2 + I_3 = \frac{V_- - V_x}{10k} - \frac{V_x}{100}$$

$\therefore I_L$  doesn't depend on  $R_L$

so for  $100 \Omega \leq R_L \leq 1 \text{ k}\Omega$ ,  $I_L = 10.1 \text{ [mA]}$

$$\frac{V_x - V_o}{R_L} = 10.1 \text{ m} \Rightarrow V_x - 10.1 \ln R_L = V_o$$

$$-1 - 10.1 \ln R_L = V_o$$

$$R_L = 100 \Omega: V_o = -2.01 \text{ [V]}$$

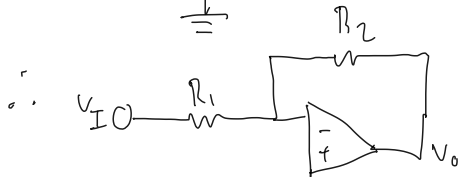
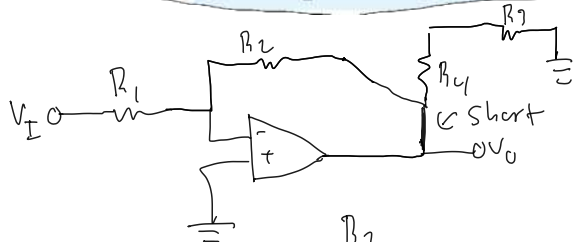
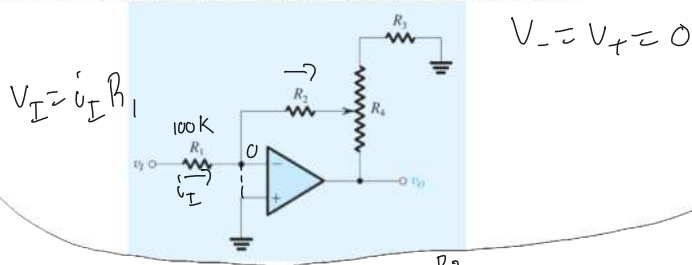
$$R_L = 1 \text{ k}\Omega: V_o = -11.1 \text{ [V]}$$

$$\frac{(-1) - (-13)}{10.1 \text{ m}} = R_L$$

$$R_L = 1188.1188 \approx 1.2 \text{ [k}\Omega\text{]}$$

max value to keep  $V_o \geq -13 \text{ [V]}$

4. Design the circuit shown in Fig. P2.35 to have an input resistance of  $100\text{ k}\Omega$  and a gain that can be varied from  $-1\text{ V/V}$  to  $-100\text{ V/V}$  using the  $100\text{-k}\Omega$  potentiometer  $R_4$ . What voltage gain results when the potentiometer is set exactly at its middle value?

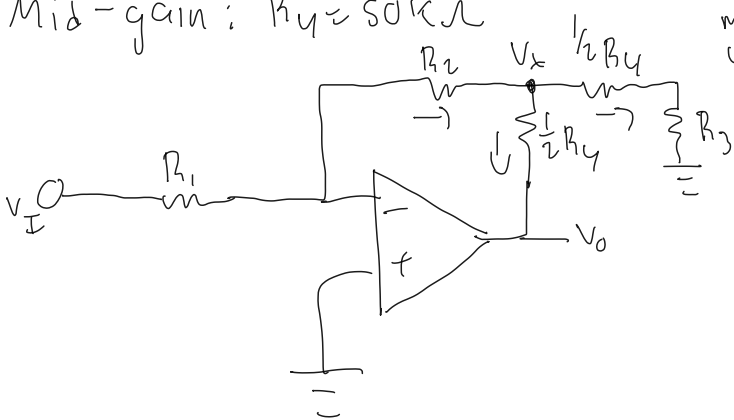


$$\frac{V_O}{V_I} = -\frac{R_2}{R_1} = -1 = A$$

$$R_2 = R_1 \quad \text{min. gain}$$

$$R_2 = 100\text{ k}\Omega$$

Mid-gain:  $R_4 = 50\text{ k}\Omega$

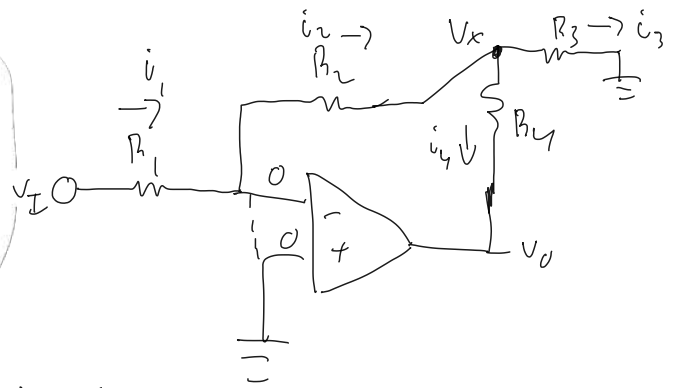


KCL at  $V_x$ :

$$-\frac{0 - V_x}{R_2} + \frac{V_x - V_O}{.5R_4} + \frac{V_x}{.5R_4 + R_3} = 0$$

$$V_x \left( \frac{1}{R_2} + \frac{1}{.5R_4} + \frac{1}{.5R_4 + R_3} \right) = V_O \left( \frac{1}{.5R_4} \right)$$

$$-V_I \frac{R_2}{R_1} \left( \frac{1}{R_2} + \frac{1}{.5R_4} + \frac{1}{.5R_4 + R_3} \right) = V_O \left( \frac{1}{.5R_4} \right)$$



$$i_1 = i_2$$

$$\frac{V_I}{R_1} = -\frac{V_x}{R_2} + \left( \frac{0 - V_x}{R_2} \right) + \frac{V_x - V_O}{R_4} + \frac{V_x}{R_3} = 0$$

$$V_x = -V_I \frac{R_2}{R_1}$$

$$V_x \left( \frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_3} \right) = V_O \left( \frac{1}{R_4} \right)$$

$$-V_I \frac{R_2}{R_1} \left( \frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_3} \right) = V_O \left( \frac{1}{R_4} \right)$$

$$\frac{V_O}{V_I} = -\left[ \frac{R_4}{R_1} + \frac{R_2}{R_1} + \frac{R_4 R_2}{R_1 R_3} \right]$$

$$\text{Since } R_2 = R_1: \frac{V_O}{V_I} = -\frac{R_4}{R_1} - 1 - \frac{R_4}{R_3}$$

$$\frac{V_O}{V_I} = -\frac{100\text{ k}\Omega}{100\text{ k}\Omega} - 1 - \frac{100\text{ k}\Omega}{R_3}$$

max gain  
since min already used

$$\frac{V_O}{V_I} = -2 - \frac{100\text{ k}\Omega}{R_3}$$

$$-100\text{ k}\Omega \approx R_3$$

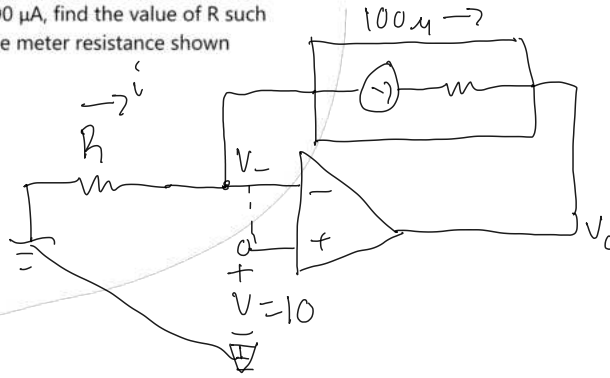
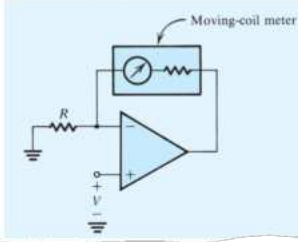
$$R_3 = 1020.41\text{ }\Omega \approx 1.02\text{ k}\Omega$$

$$\frac{V_O}{V_I} = -\left[ \frac{.5R_4}{R_1} + \frac{R_2}{R_1} + \frac{.5R_4 R_2}{R_1 (.5R_4 + R_3)} \right]$$

$$\frac{V_O}{V_I} = -\left[ \frac{50\text{ k}\Omega}{100\text{ k}\Omega} + 1 + \frac{50\text{ k}\Omega}{50\text{ k}\Omega + 1.02\text{ k}\Omega} \right]$$

$$\frac{V_O}{V_I} = -2.48$$

5. Figure below shows a circuit for an analog voltmeter of very high input resistance that uses an inexpensive moving-coil meter. The voltmeter measures the voltage  $V$  applied between the op amp's positive-input terminal and ground. Assuming that the moving coil produces full-scale deflection when the current passing through it is  $100\ \mu\text{A}$ , find the value of  $R$  such that a full-scale reading is obtained when  $V$  is  $+10\ \text{V}$ . Does the meter resistance shown affect the voltmeter calibration?



$$V_- = V_+$$

Meter resistance should not affect calibration  
Since op-amp will drive

current to maintain  $V_- = V_+ = V = 10\ \text{V}$ .

Internal resistance only affects required  $V_0$  to maintain  $100\ \mu\text{A}$  of current between  $V_- \rightarrow V_0$  (across meter).

$$\text{KCL @ } V_-: -i + 100\ \mu\text{A} = 0$$

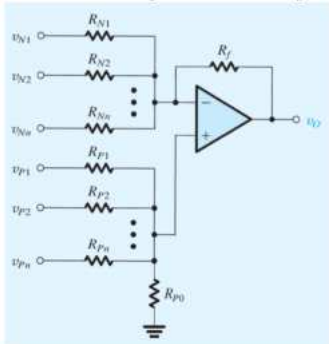
$$i = 100\ \mu\text{A}$$

$$-\frac{0 - V_-}{R} = i$$

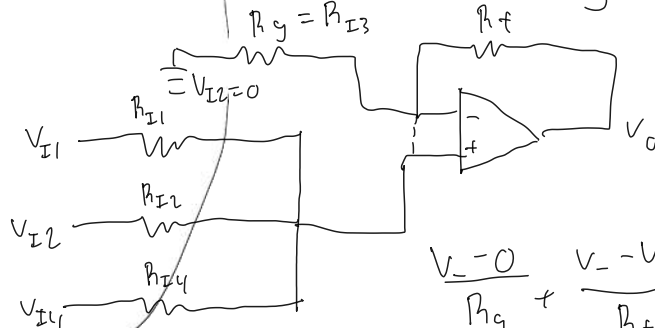
$$\frac{V_-}{i} = R$$

$$R = (100\ \text{K}\Omega)$$

6. Design a circuit, using one ideal op amp, whose output is  $v_0 = v_{i1} + 2v_{i2} - 9v_{i3} + 4v_{i4}$ . (Hint: Use a structure similar to that shown in general form in figure below)



for non-inverting



$$\frac{V_- - 0}{R_g} + \frac{V_- - V_0}{R_f} = 0$$

$$V_- \left( \frac{1}{R_g} + \frac{1}{R_f} \right) = V_0 \frac{1}{R_f}$$

$$\frac{V_0}{V_+} = \frac{V_0}{V_-} = A_+ = \left( \frac{R_f}{R_g} + 1 \right)$$

$$A_+ = 9 + 1 = 10$$

$$V_0 = \frac{V_{i1}}{R_{p1}} + \frac{V_{i2}}{R_{p2}} + \frac{V_{i4}}{R_{p4}} \quad \begin{matrix} V_{i1} = \frac{1}{A_+} = 0.1 \\ V_{i2} = 0.2 \\ V_{i4} = 0.4 \end{matrix}$$

$$\left( \frac{1}{R_{p1}} + \frac{1}{R_{p2}} + \frac{1}{R_{p4} + R_{p0}} \right)^{-1}$$

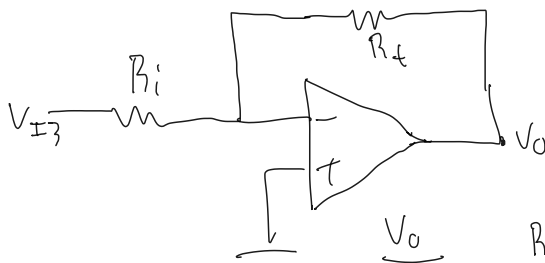
$$\left( \frac{1}{R_{p1}} + \frac{1}{R_{p2}} + \frac{1}{R_{p4} + R_{p0}} \right)^{-1} = R_f \parallel R_g = 90\text{K} \parallel 10\text{K} = 9\text{K}$$

$$\frac{1}{R_{p1}} = \frac{0.1}{9\text{K}} \quad \frac{1}{R_{p2}} = \frac{0.2}{9\text{K}} \quad \frac{1}{R_{p4}} = \frac{0.4}{9\text{K}}$$

$$R_{p1} = 90\text{K} \quad R_{p2} = 45\text{K} \quad R_{p4} = 22.5\text{K}$$

$$R_{p0} = \left( \frac{1}{R_{p1}} + \frac{1}{R_{p2}} + \frac{1}{R_{p4}} \right)^{-1} = 30\text{K}$$

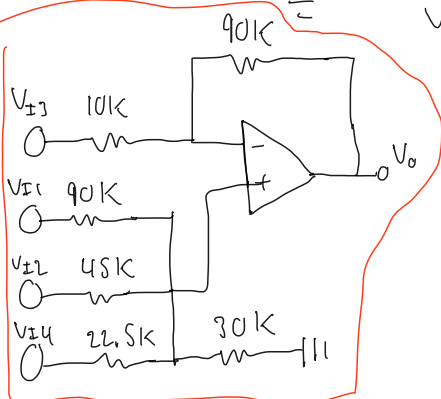
for inverting:  $\frac{V_0}{V_{i3}} = -9 = A_-$



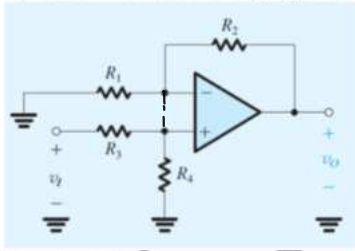
$$\frac{V_0}{V_{i3}} = -\frac{R_f}{R_i} = -9$$

$$\frac{90\text{K}}{10\text{K}} = 9 \checkmark$$

$$R_f = 90\text{K}, \quad R_{i3} = 10\text{K} = R_g$$



7. Derive an expression for the voltage gain,  $v_o/v_i$ , of the circuit below



$$V_- = V_+$$

$$\text{KCL @ } V_-: \frac{V_-}{R_1} + \frac{V_- - V_o}{R_2} = 0$$

$$V_- \left( \frac{1}{R_1} + \frac{1}{R_2} \right) = V_o \left( \frac{1}{R_2} \right)$$

$$V_- \left( \frac{R_2}{R_1} + 1 \right) = V_o$$

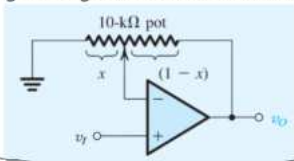
$$V_i \left( \frac{R_4}{R_3 + R_4} \right) \left( \frac{R_2}{R_1} + 1 \right) = V_o$$

$$\frac{V_o}{V_i} = \frac{R_4 R_2}{R_1 (R_3 + R_4)} + \frac{R_4}{R_3 + R_4}$$

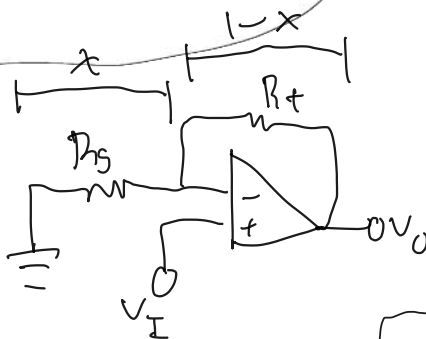
$$\frac{V_o}{V_i} = \frac{R_4 R_2 + R_1 R_4}{R_1 (R_3 + R_4)}$$

$$\frac{V_o}{V_i} = \frac{R_4}{R_1} \left( \frac{R_2 + R_1}{R_3 + R_4} \right)$$

8. The circuit below utilizes a 10-k $\Omega$  potentiometer to realize an adjustable-gain amplifier. Derive an expression for the gain as a function of the potentiometer setting  $x$ . Assume the op amp to be ideal. What is the range of gains obtained? Show how to add a fixed resistor so that the gain range can be 1 to 11 V/V. What should the resistor value be?



Non-inverting



$$R_i = x(10K)$$

$$R_f = (1-x)(10K)$$

depending on pot. wiper.

$$A = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_i}$$

$$A = 1 + \frac{(1-x)10K}{x10K}$$

$$= 1 + \frac{1-x}{x}$$

$$= 1 + \frac{1}{x} - 1$$

$$A = \frac{1}{x}$$

$$x=0: A = \frac{1}{0} = \infty$$

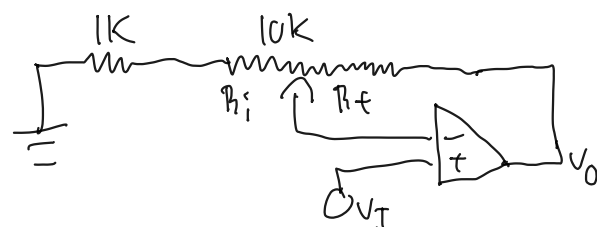
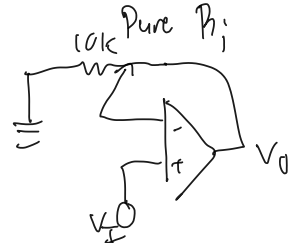
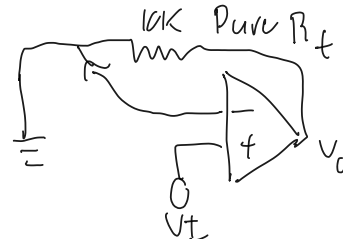
$$x=1: A = 1$$

$$A' = 1 + \frac{(1-x)10K}{x10K + R_g}$$

$$\text{max @ } x=0: A'_{\text{max}} = 1 + \frac{10K}{R_g} = 11$$

$$= \frac{10K}{R_g} = 10$$

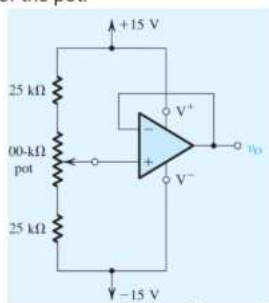
$$R_g = 1K$$



$$\therefore 1 \leq A \leq 11$$

$$\therefore 1 \leq A \leq \infty$$

9. Figure P2.59 shows a circuit that provides an output voltage  $v_o$  whose value can be varied by turning the wiper of the 100-k $\Omega$  potentiometer. Find the range over which  $v_o$  can be varied. If the potentiometer is a "20-turn" device, find the change in  $v_o$  corresponding to each turn of the pot.

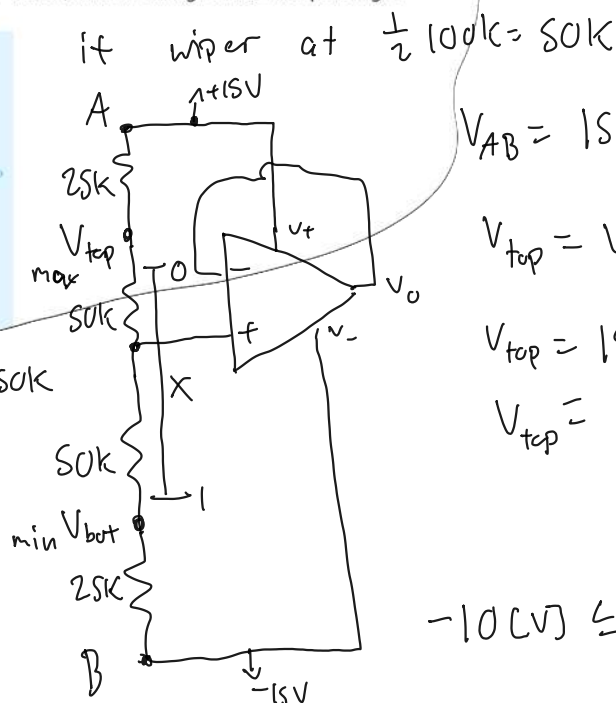


$25k + 25k + 50k = 100k$   
series

$$V_{bot} = V_A - V_{AB} \left( \frac{125k}{150k} \right)$$

$$= 15 - 30 \left( \frac{5}{6} \right)$$

$$V_{bot} = -10V$$



$$V_{AB} = 15 - (-15) = 30V$$

$$V_{top} = V_A - V_{AB} \left( \frac{25k}{150k} \right)$$

$$V_{top} = 15 - 30 \left( \frac{1}{6} \right)$$

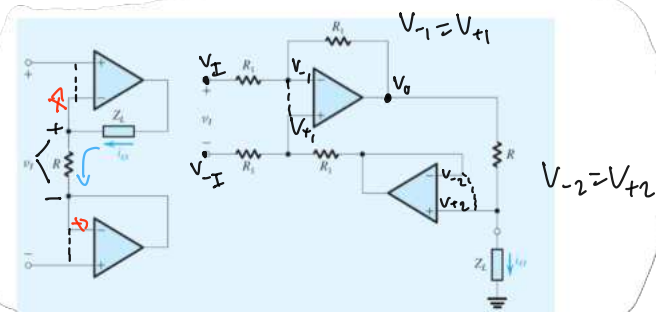
$$V_{top} = 10V$$

$-10V \leq v_o \leq 10V$  since pot. (input) ranges from -10V to 10V and  $v_- = v_+$

if  $x=0$ ,  $V_{top} = V_+ = 10V$   
if  $x=1$ ,  $V_{bot} = V_+ = -10V$   
20CV range / 20 turns =  $\boxed{1V \text{ turn}}$

10. The two circuits below are intended to function as voltage-to-current converters; that is, they supply the load impedance  $Z_L$  with a current proportional to  $v_i$  and independent of the value of  $Z_L$ . Show that this is indeed the case, and find for each circuit  $i_o$  as a function of  $v_i$ . Comment on the differences between the two circuits.

Left circuit:  $V_I = i_o R$   $i_o$  runs through  $R$   
since no current enters inputs of op-amp and  $V_I$  is voltage across  $R$  due to virtual short



Right circuit:

$$\text{KCL @ } v_{-1}: \frac{V_{-1} - V_I}{R_i} + \frac{V_{-1} - V_o}{R_f} = 0$$

$$V_{-1} = V_{+1}$$

$$V_{-1} \left( \frac{1}{R_i} + \frac{1}{R_f} \right) = \frac{V_I + V_o}{R_i}$$

$$V_{-1} \left( \frac{2}{R_i} \right) = \frac{V_I + V_o}{R_i}$$

$$V_{-1} = \frac{V_I + V_o}{2}$$

$$\text{KCL @ } v_{+1}: \frac{V_{+1} - V_I}{R_i} + \frac{V_{+1} - V_{-2}}{R_i} = 0$$

$$2V_{+1} = V_I + V_{-2}$$

$$V_{+1} = \frac{V_I + V_{-2}}{2}$$

$$\frac{V_I + V_o}{2} = \frac{V_I + V_{-2}}{2}$$

$$V_I + V_o = V_I + V_{-2}$$

$$V_{-2} = V_o + V_I - V_I = V_o - V_I$$

$$\text{KCL @ } v_{+2}: \frac{V_{+2} - V_o}{R} + \frac{V_{+2}}{Z_L} = 0$$

$$i_o = \frac{V_o - V_{+2}}{R}$$

$$V_o - V_{+2} = i_o R$$

$$V_o - (V_o - V_I) = i_o R$$

$$V_I = i_o R$$

$$\frac{V_I}{R} = i_o$$

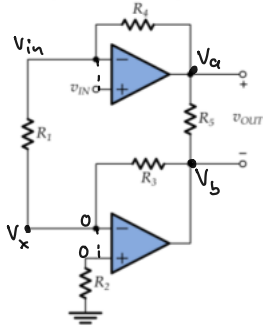
In both cases,  $i_o = \frac{V_I}{R}$

Only depends on input voltage,

not load impedance, Left circuit = non-inverting; right circuit = inverting

11. Find the gain for the following circuits, assuming ideal op-amps.

a. For this circuit,  $R_1 = R_2 = R_5 = 2 \text{ k}\Omega$  and  $R_3 = R_4 = 1 \text{ k}\Omega$ .



$$\text{KCL at } V_{in}: \frac{V_{in}}{R_1} + \frac{V_{in} - V_g}{R_4} = 0$$

$$\text{at } V_x: \frac{V_x - V_{in}}{R_1} + \frac{V_x - V_b}{R_3} = 0$$

$$V_{in} \left( \frac{1}{R_1} + \frac{1}{R_4} \right) = V_g \left( \frac{1}{R_4} \right)$$

$$-\frac{V_{in}}{R_1} - \frac{V_b}{R_3} = 0$$

$$V_{in} \left( \frac{R_4}{R_1} + 1 \right) = V_g$$

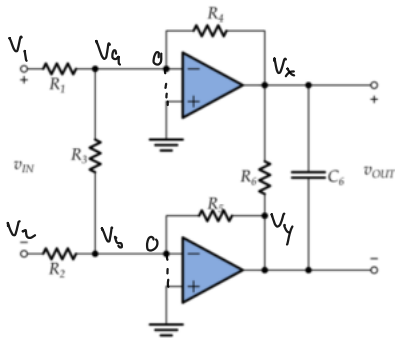
$$V_b \left( \frac{1}{R_3} \right) = -\frac{V_{in}}{R_1}$$

$$V_{out} = V_g - V_b = V_{in} \left( \frac{R_4}{R_1} + 1 \right) - \left( -V_{in} \left( \frac{R_3}{R_1} \right) \right)$$

$$V_b = -V_{in} \left( \frac{R_3}{R_1} \right)$$

$$V_{out} = V_{in} \left( 1 + \frac{R_4}{R_1} + \frac{R_3}{R_1} \right) \therefore \frac{V_{out}}{V_{in}} = \left( 1 + \frac{1\text{k}}{2\text{k}} + \frac{1\text{k}}{2\text{k}} \right) = 2$$

b. For this circuit,  $R_1 = 1 \text{ k}\Omega$ ,  $R_2 = 5 \text{ k}\Omega$ ,  $R_3 = 2 \text{ k}\Omega$ ,  $R_4 = 3 \text{ k}\Omega$  and  $R_5 = 15 \text{ k}\Omega$ ,  $R_6 = 6 \text{ k}\Omega$ , and  $C_6 = 1 \mu\text{F}$ .



$$\text{KCL at } V_g: \frac{-V_1}{R_1} - \frac{V_x}{R_4} = 0$$

$$\text{at } V_b: \frac{-V_2}{R_2} - \frac{V_y}{R_5} = 0$$

$$V_x = -V_1 \left( \frac{R_4}{R_1} \right)$$

$$V_y = -V_2 \left( \frac{R_5}{R_2} \right)$$

$$V_{out} = V_x - V_y = -V_1 \left( \frac{R_4}{R_1} \right) - \left( -V_2 \left( \frac{R_5}{R_2} \right) \right)$$

$$V_{out} = -V_1 \left( \frac{R_4}{R_1} \right) + V_2 \left( \frac{R_5}{R_2} \right)$$

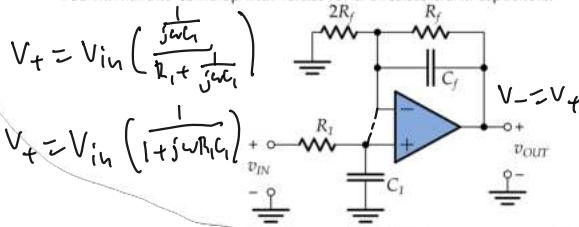
$$+V_1 - V_2 = V_{in}$$

$$= -V_{in} \left( \frac{R_4}{R_1} + \frac{R_5}{R_2} \right)$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_4}{R_1} - \frac{R_5}{R_2} = -\frac{3\text{k}}{1\text{k}} - \frac{15\text{k}}{5\text{k}}$$

$$\frac{V_{out}}{V_{in}} = -6$$

12. Find the transfer function  $H(j\omega) = V_{out}/V_{in}$  for the op amp circuit below, assuming an ideal op amp. Sketch the Bode magnitude and phase plots using straight-line approximations. You will have to come up with values for the resistors and capacitors.



$$\text{KCL at } V_-: \frac{V_-}{2R_f} + \frac{V_- - V_{out}}{R_f \parallel C_f} = 0$$

$$V_- \left( \frac{1}{2R_f} + \frac{1}{R_f + \frac{1}{j\omega C_f}} \right) = V_{out} \left( \frac{R_f + \frac{1}{j\omega C_f}}{R_f + \frac{1}{j\omega C_f}} \right)^{-1}$$

$$V_- \left( \frac{1}{2R_f} + \frac{1 + j\omega R_f C_f}{R_f} \right) = V_{out} \left( \frac{1 + j\omega R_f C_f}{R_f} \right)$$

$$V_- \left( \frac{1 + 2j\omega R_f C_f + 2}{2R_f} \right) = V_{out} \left( \frac{1 + j\omega R_f C_f}{R_f} \right)$$

$$V_- \left( \frac{3 + 2j\omega R_f C_f}{2(1 + j\omega R_f C_f)} \right) = V_{out}$$

$$V_{in} \left( \frac{1}{1 + j\omega R_1 C_1} \right) \left( \frac{3 + 2j\omega R_f C_f}{2(1 + j\omega R_f C_f)} \right) = V_{out}$$

$$\frac{V_{out}}{V_{in}} = \frac{1}{2} \left( \frac{3 + 2j\omega R_f C_f}{(1 + j\omega R_1 C_1)(1 + j\omega R_f C_f)} \right) = \frac{3}{2} \frac{(1 + \frac{2}{3}j\omega R_f C_f)}{(1 + j\omega \frac{1}{C_1 R_1})(1 + j\omega \frac{1}{R_f C_f})} = |H(j\omega)|$$

$$R_f = 10\text{k}, R_1 = 100\text{k}$$

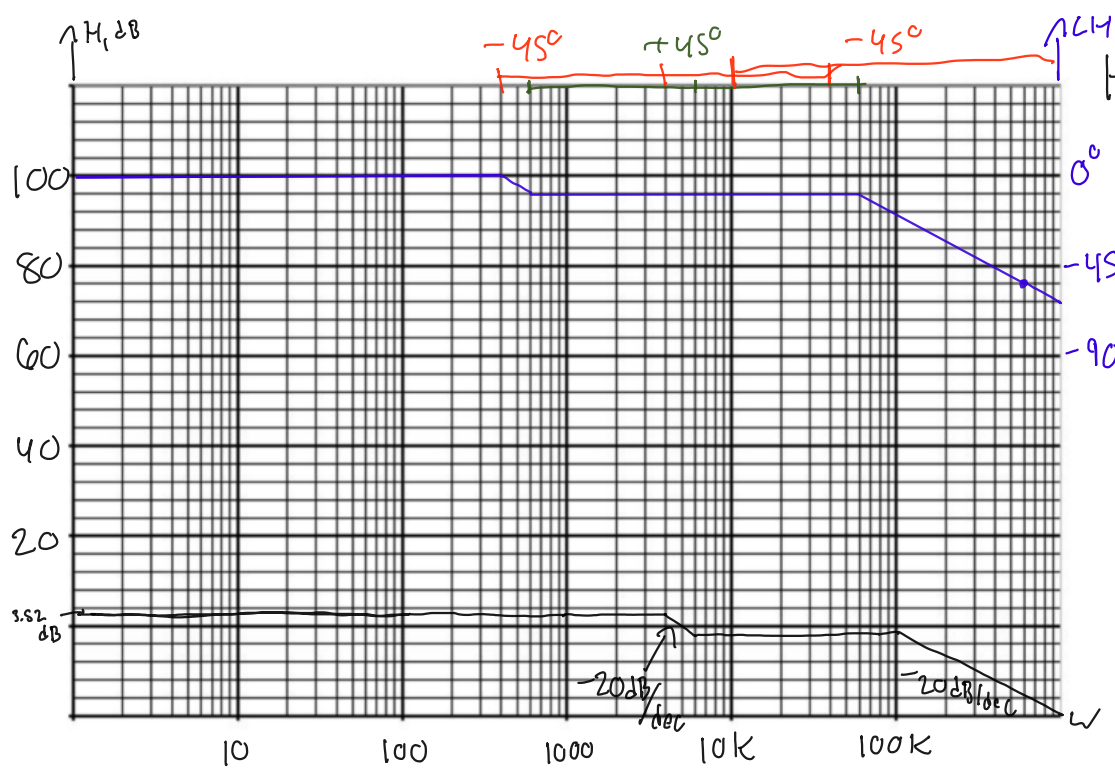
$$Z_1 = \frac{1}{R_f C_f \frac{2}{3}} = \frac{3}{2R_f C_f}$$

$$\omega = 6000 \approx \frac{3}{20\text{k} C_f} \therefore C_f = 25 \text{ nF}$$

$$P_2 = \frac{1}{R_f C_f} \therefore \omega = 4000$$

$$P_1 = \frac{1}{C_1 R_1}$$

$$\omega = 10\text{k} \approx \frac{1}{100\text{k} C_1} \therefore C_1 = 1 \text{ nF}$$



$$H(j\omega) = \frac{3}{2} \frac{(1 + j \frac{\omega}{6000})}{(1 + j \frac{\omega}{10K})(1 + j \frac{\omega}{4000})}$$

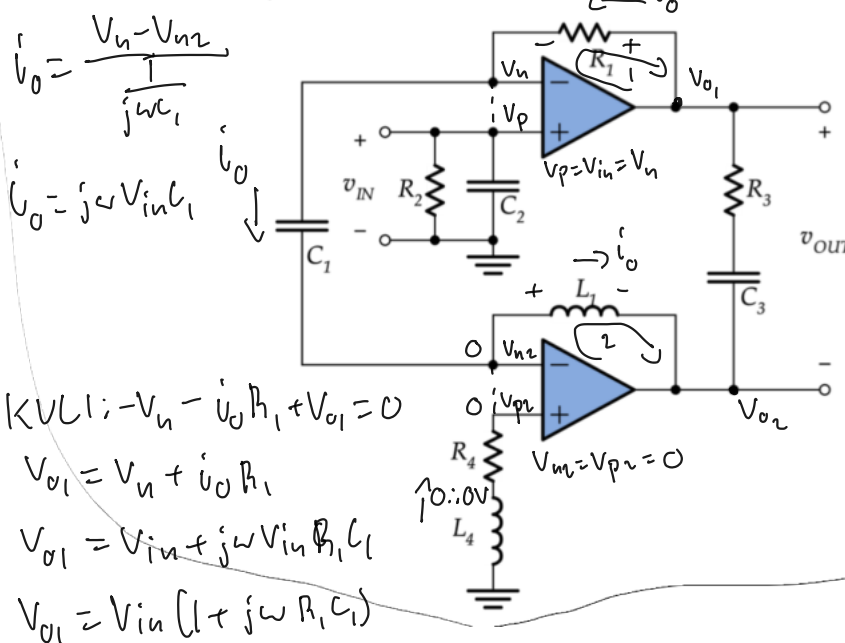
$$H(\omega=100) \approx \frac{3}{2} \approx 3.52$$

$$\angle H \approx 0^\circ$$

13. Assuming ideal op-amps,

- Find the transfer function,  $H(\omega) = V_{out}/V_{in}$
- Find the number of poles and zeros for this transfer function
- For extra credit, find the values of the poles and zeros for this transfer function

Use  $R_1=1 \text{ k}\Omega$ ,  $R_2=2.2 \text{ k}\Omega$ ,  $R_3=3.3 \text{ k}\Omega$ ,  $R_4=4.7 \text{ k}\Omega$ ,  $C_1=2.2 \text{ }\mu\text{F}$ ,  $C_2=0.1 \text{ mF}$ ,  $C_3=0.01 \text{ mF}$ ,  $L_1=3.3 \text{ mH}$ , and  $L_4=39 \text{ }\mu\text{H}$ .



$$\text{KVL 2: } -V_{out2} + i_0 j\omega L_1 + V_{out1} = 0$$

$$V_{out2} = -i_0 j\omega L_1$$

$$V_{out2} = -(j\omega V_{in} C_1) j\omega L_1$$

$$V_{out2} = \omega^2 V_{in} C_1 L_1$$

$$V_{out} = V_{out1} - V_{out2}$$

$$= V_{in} (1 + j\omega R_1 C_1 - \omega^2 C_1 L_1)$$

$$\frac{V_{out}}{V_{in}} = 1 + j\omega R_1 C_1 - \omega^2 C_1 L_1 = H(j\omega)$$

no denominator: 0 poles

2 zeros:  $z_1, z_2$

$$\text{quadratic zeros: } \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

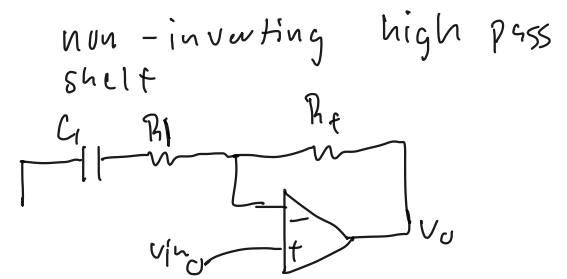
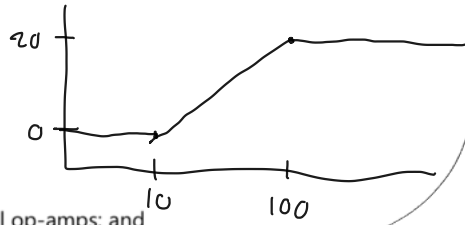
$$\frac{-(R_1 C_1) \pm \sqrt{R_1^2 C_1^2 - 4(C_1 L_1)(1)}}{2(C_1 L_1)} = \frac{-455.2 - 302.575 \cdot 10^3}{2 \text{ zeros}}$$

14. Design a circuit that meets the following specifications using only parts from the list below (you do not have to use all of the parts).

- Gain is 20 dB  $\pm$  6 dB for  $\omega \geq 100$  rad/sec
- Gain is 0 dB  $\pm$  6 dB for  $\omega < 10$  rad/sec
- $R_{in} > 150$  k $\Omega$

Available parts:

- (10) resistors of any value  $< 10$  k $\Omega$ ; • (5) ideal op-amps; and
- any capacitors.



$$H(j\omega) = \frac{1 + j\omega(R_1 + R_f)C_1}{1 + j\omega R_1 C_1} = \frac{1 + j\frac{\omega}{\frac{1}{(R_1 + R_f)C_1}}}{1 + j\frac{\omega}{\frac{1}{R_1 C_1}}}$$

$$P_1 = \frac{1}{R_1 C_1} = 100 \therefore C_1 = 10 \mu F$$

$$Z_1 = \frac{1}{C_1(R_1 + R_f)} = 10$$

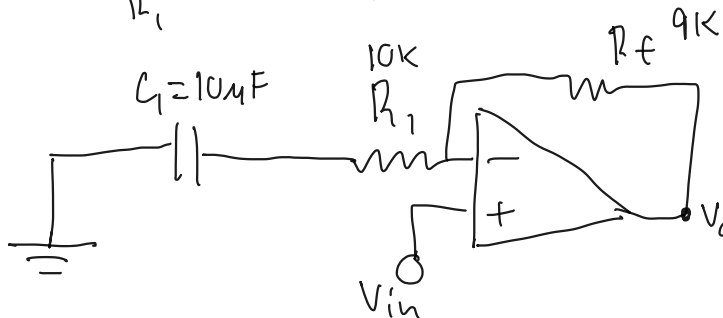
$$\omega \rightarrow 0 \quad H \approx 1 \quad 20 \log(1) = 0$$

$$100 \leq \omega \rightarrow \infty \quad H \approx 1 + \frac{R_f}{R_1} \quad \text{or} \quad 1 + \frac{R_f}{R_1} = 20 \text{ dB} = 20 \log(10)$$

$$1 + \frac{R_f}{R_1} = 10$$

$$\frac{R_f}{R_1} = 9 \therefore R_f = 9 \text{ k} \leq 10 \text{ k} \checkmark$$

$$R_1 = 1 \text{ k} \leq 10 \text{ k}$$



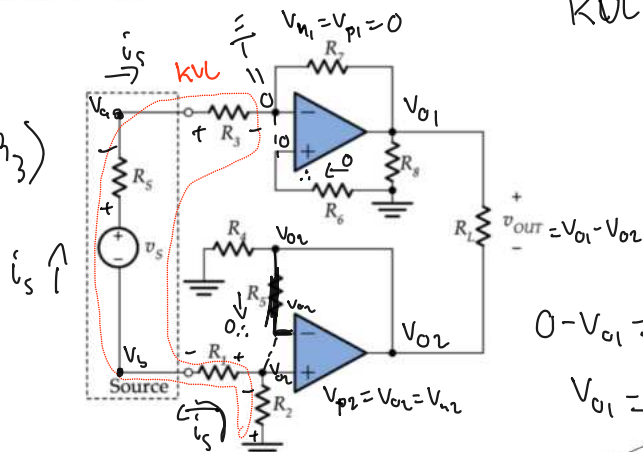
15. Assuming ideal op-amps,

- Find the voltage gain,  $v_{out}/v_s$
- Find the input resistance seen by the source

Use  $R_5=6.9\text{ k}\Omega$ ,  $R_1=3.3\text{ k}\Omega$ ,  $R_2=4.7\text{ k}\Omega$ ,  $R_3=2.2\text{ k}\Omega$ ,  $R_4=75\text{ k}\Omega$ ,  $R_5=19\text{ k}\Omega$ ,  $R_6=27\text{ k}\Omega$ ,  $R_7=86\text{ k}\Omega$ ,  $R_8=10\text{ k}\Omega$ , and  $R_L=5.6\text{ k}\Omega$ .

$$V_a - 0 = i_s R_3$$

$$V_a = \frac{V_s}{17.1\text{ k}} (R_3)$$



$$\text{KVL: } i_s R_2 + i_s R_1 - V_s + i_s R_5 + i_s R_3 = 0$$

$$V_s = i_s (R_1 + R_2 + R_3 + R_5)$$

$$i_s = \frac{V_s}{17.1\text{ k}}$$

$$0 - V_{o1} = i_s R_7$$

$$V_{o1} = -i_s R_7 = -\frac{V_s}{17.1\text{ k}} R_7$$

$$0 - V_{o2} = i_s R_2$$

$$V_{o2} = -i_s R_2 = -\frac{V_s}{17.1\text{ k}} R_2$$

$$0 - V_b = i_s (R_2 + R_1)$$

$$V_b = i_s (R_1 + R_2)$$

$$= \frac{V_s}{17.1\text{ k}} (R_1 + R_2)$$

$$V_{out} = -\frac{V_s}{17.1\text{ k}} R_7 - \left( -\frac{V_s}{17.1\text{ k}} R_2 \right)$$

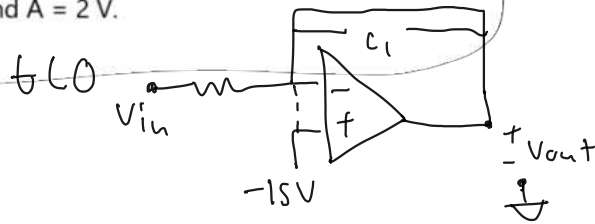
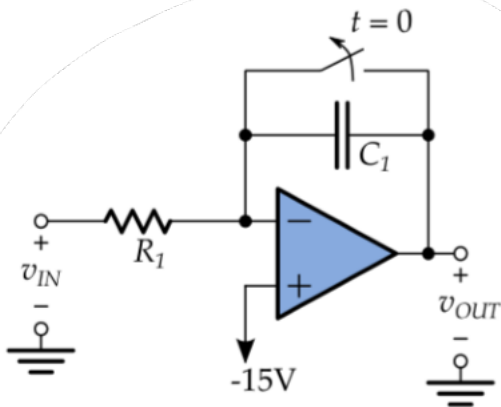
$$= V_s \left( -\frac{R_7}{17.1\text{ k}} + \frac{R_2}{17.1\text{ k}} \right)$$

$$\frac{V_{out}}{V_s} = -4.75$$

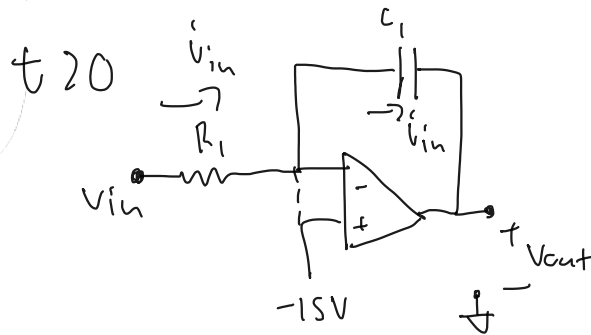
Resistance seen by source:  $V_s = i_s R_s$

$$R_s = \frac{V_s}{i_s} = 17.1\text{ k}\Omega$$

16. The op-amp in the circuit below is ideal. The switch is closed until  $t = 0$ , at which time it opens and stays open. Find  $v_{OUT}(t)$  for the range  $-10 \text{ s} < t < 10 \text{ s}$ . Use  $R_1 = 2 \text{ k}\Omega$ ,  $C_1 = 3 \text{ }\mu\text{F}$ , and  $v_{IN}(t) = A \sin(\omega_0 t)$ , where  $\omega_0 = 300 \text{ rad/s}$  and  $A = 2 \text{ V}$ .



$$v_{OUT}(0^-) = -15 \text{ V} = v_{OUT}(0^+)$$



$$\dot{v}_{in} = \dot{v}_C = C \frac{dv_C}{dt}$$

$$v_C = \frac{1}{C} \int \dot{v}_C dt + v_{OUT}(0)$$

$$\text{KCL @ } v_-: \frac{v_{in} - v_-}{R_1} + C_1 \frac{d}{dt} (-15 - v_{OUT}) = 0$$

$$\frac{v_{in} + 15}{R_1} = - \left[ C_1 \frac{d}{dt} (-15 - v_{OUT}) \right]$$

$$\frac{v_{in} + 15}{R_1 C_1} = - \frac{d}{dt} (-v_{OUT})$$

$$\frac{1}{R_1 C_1} \left[ \int_0^t v_{in} + 15 dt \right] = v_{OUT} \Big|_0^t$$

$$\frac{1}{R_1 C_1} \left[ \int_0^t 2 \sin(300t) + 15 dt \right] = v_{OUT}(t) - v_{OUT}(0)$$

$$\frac{1}{R_1 C_1} \left[ -\frac{2}{300} \cos(300t) + 15t \Big|_0^t \right] = v_{OUT}(t) + 15$$

$$-\frac{10}{9} \cos(300t) + 2500t - \left[ -\frac{10}{9} \cos(0) + 0 \right] = v_{OUT}(t) + 15$$

$$v_{OUT}(t) = -\frac{10}{9} \cos(300t) + 2500t - \frac{125}{9} \quad 0 \leq t \leq 10$$